

## RESEARCH ARTICLE OPEN ACCESS

# Seeing the Child Within: A Randomized Controlled Trial of Reflective Functioning Intervention for Parents of Autistic Children

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## ABSTRACT

Parents of autistic children often face elevated emotional demands and challenges in interpreting their child's inner experience. Parental reflective functioning (PRF), the capacity to understand behavior in terms of underlying mental states, is considered a key mechanism in fostering emotionally attuned and adaptive caregiving. Despite its importance, few interventions directly target PRF, and fewer still have tested brief, group-based formats designed for broad accessibility. We conducted a randomized controlled trial to evaluate a brief, virtual PRF intervention delivered via Zoom to 249 parents of autistic children. Participants were randomly assigned to either an intervention group ( $n = 129$ ) or a waitlist control group ( $n = 130$ ), and completed assessments at baseline and postintervention. PRF was measured using a standardized interview, while self-report questionnaires assessed parenting self-efficacy, beliefs about emotional malleability, depressive symptoms, and child adaptive functioning. Generalized estimating equations revealed significant time-by-group interactions favoring the intervention group across several outcomes. Relative to controls, intervention participants reported greater gains in reflective functioning, parenting self-efficacy, and emotional beliefs, alongside improvements in children's communication and conceptual adaptive functioning. Effects were most pronounced among parents of younger children. Within the intervention group, additional follow-up assessments indicated sustained benefits on workshop-specific outcomes. Findings support the feasibility and efficacy of brief, scalable PRF-based programs for caregivers of autistic children. To our knowledge, this is the first randomized trial to demonstrate that a short, online, parent-focused intervention can enhance reflective parenting and promote emotional and relational benefits in this population.

## 1 | Introduction

Raising a child with autism, a neurodevelopmental condition characterized by repetitive behaviors and difficulties in social communication, presents families with unique challenges (Aviad et al. 2024). Parents frequently experience elevated stress, emotional exhaustion, and a sense of uncertainty as they navigate their child's neurodiverse communication patterns,

sensitivities, and behaviors (Lievore et al. 2024). These stressors can compromise parents' emotional regulation capacities, undermine their sense of parental efficacy, and strain the parent-child relationship: essential for children's emotional security and long-term developmental success (Higgins et al. 2023). As parents shape their child's environment, supporting parental well-being and regulatory capacities is critical for fostering adaptive outcomes in autistic children.

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## 1.1 | Parental Emotion Regulation (ER) and Reflective Functioning

Parental ER is central to effective parenting (Aydin 2023; Enav et al. 2019). ER enables parents to manage their emotional responses, maintain composure during difficult interactions, and model adaptive coping strategies for their children (Aviad et al. 2024; Malberg 2021). Difficulties in ER can undermine parental self-efficacy (PSE), elevate parental stress levels, and impair parents' abilities to remain emotionally available to their children (Aviad et al. 2024).

Parental reflective functioning (PRF), the parental capacity to understand and interpret the child's behavior in terms of underlying mental and emotional states (Slade 2005), supports effective parental ER by reducing emotional activity, promoting empathy, and guiding more regulated responses (Enav et al. 2022). PRF also serves as a regulatory resource during emotionally challenging interactions (Enav et al. 2022). Stronger PRF is associated with emotional availability, reduced parenting stress, and more adaptive functioning in children (Enav et al. 2019; Huynh et al. 2024; Luyten et al. 2017). PRF and ER form a dynamic system that powers responsive parenting (Luyten et al. 2017; Rutherford et al. 2015) and are especially critical for fostering resilience in families with autistic children.

## 1.2 | PRF and Autism

PRF is critical in sustaining emotionally attuned and developmentally supportive caregiving among parents of autistic children (Slade 2009). Autistic children often express themselves in inconsistent or ambiguous ways, which could be due to limited verbal language, reduced social referencing, or heightened sensory reactivity (Griffin et al. 2022; Mazefsky and White 2014). This complicates parents' efforts to understand their child's needs, causing PRF to feel more difficult and emotionally taxing (Fonagy and Allison 2021).

Therefore, parents of autistic children frequently report reduced PRF, lower sense of parental efficacy, greater stress, and greater difficulty maintaining psychological flexibility compared to parents of neurotypical children (Brennan et al. 2024). This underscores the importance of targeted interventions that strengthen PRF, especially under stressful conditions. Enhanced PRF supports parental well-being and more effective parenting, promoting resilience and improved developmental trajectories in children (Enav et al. 2019; Huynh et al. 2024; Luyten et al. 2017; Slade 2009).

## 1.3 | PRF, PSE, Mental Health, and Parental Beliefs

Parents' internal belief systems and mental health also impact reflective and regulated caregiving. Parents' implicit theories of emotions (ITE) guide parents' beliefs about whether emotions are fixed or malleable (Tamir et al. 2007). Parents who view emotions as dynamic are more likely to engage in self-reflection and maintain emotional flexibility during stressful moments (Luyten et al. 2017; Tamir et al. 2007). Believing in emotional

flexibility may enhance PRF as parents stay curious about their child's mental states, especially those difficult to decipher.

Closely linked to ITE is PSE, the caregiver's confidence in his or her ability to manage parenting challenges. High PSE is associated with less stress, improved mental health, and more adaptive child functioning in families of autistic children (Almendingen and Pilkington 2024; Brennan et al. 2024). Low PSE may undermine PRF by increasing parental reactivity, discouraging help-seeking behaviors, and reinforcing negative interaction patterns (Hamama 2022; Karst and Van Hecke 2012). Interventions that promote reflective insight and emotional understanding may enhance PSE and PRF as parents feel more capable of interpreting and responding to their child's needs (Enav et al. 2019, 2022).

Underlying both PRF and PSE is the broader foundation of mental health. Caregivers of autistic children face elevated risks of depression, anxiety, and burnout (Hamama 2022). This may reduce the parent's emotional availability and interfere with PRF (Enav et al. 2019). Supporting parental psychological health is not only a goal in itself but is also a condition necessary to the successful cultivation of reflective, regulated parenting, thereby improving outcomes in autistic children.

## 1.4 | Child Behavioral Outcomes

Autistic children frequently present with a range of behavioral and emotional difficulties (Mazefsky and White 2014) that often stem from underlying neurodevelopmental differences and can be exacerbated by the caregiving environment (Neece et al. 2019). For instance, if parents misinterpret child behaviors as defiant or threatening, a more emotionally reactive response from the parent tends to be exhibited. This cycle of misunderstanding exerts strain on the parent-child relationship (Hastings 2002); PRF plays a pivotal role in breaking this cycle. For instance, parents are more likely to respond calmly and consistently when they reflect on the mental and emotional states driving their child's behaviors (Enav et al. 2022). This contributes to improved child outcomes like reduced externalizing problems, more effective self-regulation, and enhanced emotional security (Enav et al. 2022; Malberg 2021).

When PRF is compromised, parents may struggle to interpret their child's behaviors, leading to misaligned responses that escalate negative interaction patterns and contribute to relational and behavioral deterioration (Camoirano 2017; Gross 2015). Thus, interventions that bolster PRF and ER do more than support parental functioning; they have the potential to reshape the parent-child dynamic in ways that promote healthier developmental and behavioral trajectories for autistic children.

## 1.5 | PRF Interventions for Parents of Autistic Children

Several intervention models have focused on enhancing PRF in families of autistic children. Interventions that focus on enhancing PRF in families of autistic children vary in delivery format and theoretical emphasis but share a common goal of strengthening parental insight, responsiveness, and emotional

availability. For instance, Azimi et al. (2024) evaluated a group-based intervention that integrated PRF with cognitive reframing strategies. Their findings demonstrated gains in PRF and notable reductions in child aggression, highlighting the potential for combining emotional and cognitive frameworks. Similarly, Juffer et al. (2018) developed VIPP-AUTI, a home-based video-feedback intervention adapted for families of autistic children. This model emphasized reflective parenting and led to improvements in both parental insight and parent-child joint attention: key social-communicative capacities in autism.

Yet there still exists a gap in the literature. Since the beginning of COVID-19, remote online interventions have become increasingly common (Probst et al. 2021). Online interventions grant one the ability to participate regardless of distance, presence of pandemic, or war. Further, the flexibility offered by online interventions is especially valuable to parents with autistic children. Depending on the support needs of their children, parents may be unable to attend traditional in-person interventions owing to work, childcare, or other commitments associated with caring for an autistic child (Hermaszewska and Sin 2021).

Bearing this in mind we expanded upon a previous successful intervention by Enav et al. (2019). This intervention was a brief, group-based, in-person pilot program that yielded gains in PRF, ITE, PSE, and the child's emotional and behavioral functioning. Participants reported increased reflective insight and confidence in parenting. Parents also reported reductions in difficulties pertaining to their children, particularly those with higher levels of baseline behavioral symptoms (Enav et al. 2019). The current study builds on this foundational work, aiming to address its limitations and expand its reach. Specifically, we test the Enav et al. (2019) model in a randomized, fully online format, improving accessibility while rigorously evaluating changes in PRF, ITE, PSE, and the child's adaptive functioning. This design allows for a more robust examination of the intervention's underlying mechanisms and its potential scalability in diverse caregiving contexts. Specifically, while all the aforementioned parental variables are closely interconnected, the current study conceptualizes PRF as a central mechanism through which changes in parental emotional functioning may influence child outcomes. Improvements in parents' ER capacities and emotional awareness are expected to enhance parents' ability to mentalize and respond reflectively to their child's emotional experiences. In turn, stronger PRF may promote greater PSE, more attuned parent-child interactions, and improvements in children's adaptive functioning. This proposed pathway distinguishes between proximal parent-level changes and more distal child-related outcomes, providing a clearer theoretical framework for understanding the intervention's mechanisms of change.

## 1.6 | The Current Study

The current study evaluates a brief, manualized, online group-based parenting intervention grounded in emotion-focused and PRF-informed principles exemplified by a previous intervention by Enav et al. (2019). While prior pilot data demonstrated promising results, they were limited by small, homogenous samples. The current study extends the pilot by possessing a larger, heterogeneous sample. The current study employs a randomized controlled trial with a waitlist control design and fully remote

delivery, thereby enhancing both methodological rigor and accessibility for families of autistic children.

To the best of our knowledge this is the first RCT to evaluate a brief, online, PRF-based parenting intervention for parents of autistic children. We hypothesized that in comparison to the control group, parents in the intervention group would demonstrate: (1) increased PRF, (2) greater PSE, (3) more malleable beliefs about ER, (4) lower levels of depressive symptoms, and (5) improvements in the child's adaptive functioning as based on parental reports. This study was registered in the clinical trial system at [clinicaltrials.gov](https://clinicaltrials.gov) (registration number: NCT06598605).

## 2 | Methods

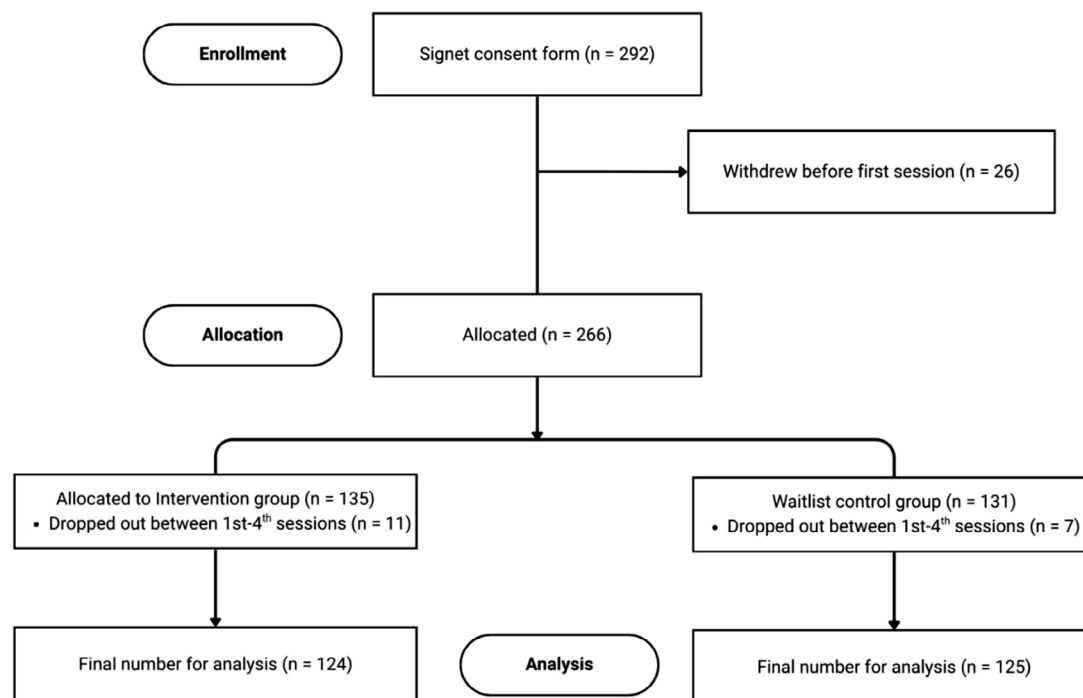
Our study was advertised on social media platforms such as Facebook and WhatsApp. Interested parents left contact information and parents were called by one of two research assistants (RAs) on the team. Parents were given more information about the project and the RA assessed parents' eligibility to participate. The study's design is an RCT parallel-group, waitlist-control trial with a roughly equal allocation ratio ( $\pm 1:1$ , 129 intervention:130 control). To receive a minimum power of 0.85 for the point data taken at two points in time, we used the Diggle (2002) power analysis framework to give a small to medium effect size ( $ES=0.20$ ),  $\alpha=05$ , and predicted intraclass correlation coefficient (ICC) of 0.15. This yielded a required sample size of  $N=207$ , yet we inflated this number by a possible attrition rate of 15%. Altogether, we set 244 as the minimum number for subjects in the study. This overall number was later randomly divided into two groups for the intervention program. The study was reported in accordance with the CONSORT 2010 guidelines.

### 2.1 | Statistical Analyses

To assess changes in child developmental indicators and parental emotional and cognitive outcomes over time, we employed generalized estimating equations (GEE), an extension of the generalized linear model designed to account for correlated repeated measurements across time points. GEE was selected because it provides population-averaged estimates while accounting for within-subject correlations and offers robust standard errors with fewer assumptions regarding data distribution. Compared to mixed linear models (MMLM), GEE provides a more parsimonious and computationally efficient approach for repeated-measures analyses focused on population-level effects rather than subject-specific trajectories.

An unstructured working correlation matrix was specified, reflecting no a priori assumptions regarding the pattern of correlations between repeated observations. Statistical significance was evaluated using Wald  $\chi^2$  tests. Effect size estimates were derived from Wald  $\chi^2$  statistics and are reported as Cramér's  $V$  coefficients, following recommendations for GEE-based analyses (Rosenberg 2010). Intention-to-treat principles were not applied to these analyses.

For certain workshop-related assessments, two additional follow-up measurements were conducted at T3 and T4. These



**FIGURE 1** | CONSORT flow diagram.  $N=292$  respondents initially enrolled in the study but 26 withdrew, resulting in  $N=266$  participants to be allocated. After allocation and dropouts, the final number of participants was  $N=124$  for the intervention group and  $N=125$  for the waitlist control group.

measures were obtained from the intervention group and not the control as the latter shed its status once it obtained the intervention. Transforming the control group into the waitlist group was deemed ethically necessary to not deprive struggling parents of an effective intervention. The Reliable Change Index (RCI; Bauer et al. 2004) was calculated for all individuals, which was the corrected change between the postintervention and preintervention measurements, where corrected means that both the preintervention and the internal consistency were integrated in the individual's time difference calculation.

Group (intervention vs. control), time, and their interaction are fixed effects in each model. The significance of main and interaction effects was tested using Wald's chi-square test. For main effects we used a rejection criterion of  $p < 0.05$  and for interaction terms we adopted an exploratory threshold of  $p < 0.10$ . We examined predicted subgroup means where significant interactions were detected. Subgroup means are graphically presented using bar charts with Latin letter annotations to denote statistically ranked differences between means. Additionally, due to developmental differences in ABAS scoring systems, we analyzed younger children (ages 1–5) and older children (ages 6–18) separately in analyses of adaptive behavior outcomes.

## 2.2 | Participants

The final sample included 249 parents of children diagnosed with autism. An additional 46 parents provided informed consent but did not complete the study procedures. Of these, 28 participants withdrew prior to allocation and 18 discontinued

following allocation to study condition, including 11 participants from the intervention group and seven from the waitlist control group (see Figure 1). When asked about their reasons for discontinuing participation, parents primarily reported scheduling and timing difficulties. Concerns related to emotional strain, fatigue, or exhaustion were also identified as contributing factors. Because questionnaire completion in Qualtrics required responses prior to submission, the final analytic dataset did not contain missing questionnaire data.

Participants were randomly allocated to either group through the use of a blinded randomization feature on Excel using the “RAND” function. The same two RAs worked on allocation. Allocation to study condition was conducted independently, and interview coders and outcome assessors were blinded to participants' group assignment. Participants assigned to the waitlist control condition were offered the intervention following completion of the control assessment period. Outcome assessors and PDI coders remained blind to participants' allocation throughout the study.

Regarding participant demographics, 194 children were male (77.9%) and 55 were female (22.1%). Gender distribution was similar across the intervention and control groups with no significant differences observed ( $\chi^2=0.53$ , ns). Children's ages ranged from 12 to 216 months ( $M=94.5$ ,  $SD=47.2$ ), corresponding to 1–18 years of age. Parents' ages ranged from 29 to 62 years ( $M=42.2$ ,  $SD=6.5$ ). Group comparisons revealed no significant differences at baseline in child age, parent age, number of children, income level, or education, indicating that randomization was successful in balancing key sociodemographic variables between groups (see Table 1).

**TABLE 1** | Baseline demographic characteristics of participants in control and intervention groups.

|                                          | Control  | Control       | Intervention | Intervention  |          |
|------------------------------------------|----------|---------------|--------------|---------------|----------|
|                                          | <i>N</i> | Mean (SD)     | <i>N</i>     | Mean (SD)     | <i>t</i> |
| Child age (months)                       | 125      | 94.59 (44.97) | 124          | 94.45 (49.54) | 0.02     |
| Parents' age (years)                     | 125      | 42.56 (6.67)  | 124          | 41.80 (6.33)  | 0.93     |
| Number of children                       | 125      | 2.46 (1.00)   | 124          | 2.33 (1.17)   | 0.97     |
|                                          |          |               |              |               |          |
|                                          |          |               | Percent      | Percent       |          |
| Parent gender                            |          |               |              |               | 0.00     |
| Male                                     |          | 11            | 8.8          | 11            | 8.9      |
| Female                                   |          | 114           | 91.2         | 113           | 91.1     |
| Child gender                             |          |               |              |               | 0.53     |
| Male                                     |          | 95            | 76.0         | 99            | 79.8     |
| Female                                   |          | 30            | 24.0         | 25            | 20.2     |
| Income                                   |          |               |              |               | 0.58     |
| Below average                            |          | 29            | 23.2         | 27            | 22.0     |
| Average                                  |          | 50            | 40.0         | 55            | 44.7     |
| Above average                            |          | 46            | 36.8         | 41            | 33.3     |
| Education level                          |          |               |              |               | 5.72     |
| Up to 12 years of education              |          | 6             | 4.8          | 2             | 1.6      |
| 12 years (a partial high school diploma) |          | 4             | 3.2          | 6             | 4.8      |
| 12 years with full high school diploma   |          | 14            | 11.2         | 7             | 5.6      |
| Postsecondary diploma studies            |          | 18            | 14.4         | 15            | 12.1     |
| Academic—undergraduate                   |          | 47            | 37.6         | 52            | 41.9     |
| Academic—master's degree or higher       |          | 36            | 28.8         | 42            | 33.9     |

Note: Two-independent sample *t*-tests were used for continuous variables; Pearson's chi-square tests ( $\chi^2$ ) were used for categorical comparisons. All "between-group" differences were nonsignificant ( $p > 0.05$ ).

## 2.3 | Procedure

### 2.3.1 | Recruitment and Screening

Participants were recruited through word of mouth and social media platforms like Facebook and WhatsApp. Interested parents completed an online contact form and were contacted by a trained RA who explained the study goals and procedures. Participation was completely voluntary and free of charge. Eligibility was assessed via a screening call which confirmed that participants were (1) fluent in Hebrew, (2) had an autistic child between the ages of 2–18, (3) who demonstrated moderate-to-low support needs, and (4) ensured that participants had access to Zoom. Preliminary verbal consent was obtained at this stage.

### 2.3.2 | Randomization and Consent

Eligible families were randomly sorted into either the intervention or waitlist control group using an Excel-based allocation procedure. Each group received a condition-specific consent

form outlining study procedures. Parents in the intervention group received a 100 NIS ( $\pm 33$  USD) gift card while parents in the control group received a 150 NIS ( $\pm 50$  USD) gift card to account for the additional assessment time (the control group completed questionnaires at three time points, while the intervention group completed two).

Parents were asked to submit formal documentation of their child's autism diagnosis which was then securely stored in a password-protected Google Drive folder. Each participant was assigned a unique ID code and entered into a secure tracking system. Personal identifiers were stored separately from study data and deleted upon completion of data processing. Data access was restricted to authorized research team members and all analyses and reporting were conducted anonymously.

### 2.3.3 | Assessment Procedures

All participants completed a Zoom-based interview prior to completing questionnaires. The primary interview measure was



the PDI-R2-Short Form, a semistructured interview designed to assess PRF toward the child (Slade et al. 2003). Interviews lasted approximately 45 min and were conducted by trained graduate students under the close supervision of Dr. Enav. All interviewers had completed CITI ethics certification.

The timing of assessments varied by group allocation. Parents in the intervention group completed interviews and questionnaires at two time points: preintervention and postintervention. Parents in the control group completed two assessments spaced 1 month apart. Zoom interviews were audio-recorded for analysis and all video recordings were immediately deleted to ensure participant privacy. Questionnaires were administered using Qualtrics.

### 2.3.4 | Intervention Delivery

The intervention was a brief, manualized parenting workshop adapted from Enav et al. (2019). It consisted of four weekly 90-min sessions delivered via Zoom and focused on ITE, ER, and PRF. Workshops were facilitated by three licensed clinical psychologists who were trained and supervised by a senior clinical psychologist with extensive experience working with children and parents.

To promote intervention fidelity, all sessions were delivered according to a structured treatment manual developed by the research team, which specified session goals, discussion themes, reflective exercises, and facilitation procedures. Facilitators remained consistent throughout the study and attended weekly supervision meetings that included review of session implementation, discussion of clinical and logistical challenges, case consultation, and monitoring of adherence to the intervention protocol. Any participant concerns or deviations from the protocol were discussed and addressed collaboratively during supervision in order to maintain consistency across groups and facilitators.

Parents were required to attend at least three Zoom sessions with their camera turned on in order to remain in the intervention. Each session combined psychoeducation, guided discussion, role-play, and homework (see Enav et al. 2019). At the end of each session, participants received weekly session summaries. Session 1 introduced the nature of emotions, the concept of ER, and adaptive versus maladaptive strategies. Session 2 focused on PRF, highlighting its role in understanding a child's feelings, intentions, and behaviors. Sessions 3 and 4 deepened skills through discussion of real-life challenges, role-play of homework examples, and collaborative problem-solving. Homework between sessions encouraged reflection on both parents' and children's emotional experiences in addition to the application of workshop strategies in everyday life.

### 2.3.5 | Postintervention and Follow-Up

Participants completed postintervention interviews and questionnaires following the final session. Parents were also invited to participate in follow-up assessments at 3- and 6-months postintervention. These follow-ups included the Parenting Sense of Competence Scale, the ITE scale, and general open-ended questions regarding satisfaction with the intervention. Each follow-up assessment required approximately 30 min to complete, and parents were compensated 25 NIS per questionnaire. As participants

in the waitlist control group were subsequently offered the intervention, no control group was available for comparison at the follow-up stages. Therefore, longitudinal analyses only reflect within-group change; causal inferences at follow-up are limited.

## 2.4 | Ethical Considerations

The study was approved by the Ethics Committee of the Faculty of Education at the University of Haifa (Approval No. 491/22). A centralized tracking system was used to manage and document all budget allocations and gift card distributions.

## 2.5 | Measures

### 2.5.1 | PRF

PRF was assessed using the Parent Development Interview—Revised Short Form (Slade et al. 2003), a semistructured interview consisting of 15 open-ended questions and lasting approximately 45 min. The interview explores how parents understand their child's thoughts, feelings, and intentions in emotionally significant moments. A sample question is "Can you give me an example from the past week when you and your child really clicked?" Responses are scored on a 1–7 Likert scale, with an overall score denoting the parent's reflective ability. The PDI also probes the parent's view of the parent–child relationship. The PDI was administered in Hebrew and has been validated in previous studies (Dollberg and Hanetz-Gamliel 2023; Gur et al. 2023).

Coding was conducted using the Addendum to the Reflective Functioning Scoring Manual (Slade et al. 2004). Two trained raters were involved in coding the PDI interviews. To assess interrater reliability, a subset of 50 interviews was independently coded by both raters. Interrater agreement was high (Pearson's  $r = 0.96$ ,  $p < 0.001$ ). Cases in which ratings differed by more than one point were reviewed during supervision meetings and discussed until consensus was reached.

### 2.5.2 | ITE

The ITE scale (Cronbach's  $\alpha = 0.95$ ; Tamir et al. 2007) was used to assess participants' beliefs about the malleability of emotions. The scale includes four statements representing different views on how much emotions can be changed (e.g., "Everyone can learn to control their emotions"). Participants rated their agreement with each statement on a 5-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*).

### 2.5.3 | PSE

PSE was assessed using the Parenting Sense of Competence scale (PSOC; Gibaud-Wallston and Wandersman 1978), a 17-item self-report questionnaire that evaluates parents' confidence in their caregiving abilities. Items assess two core domains: value/comfort (e.g., "Being a parent makes me tense and anxious") and skills/knowledge (e.g., "I meet my own personal expectations for expertise in caring for my child"). Responses were

rated on a 6-point Likert scale ranging from 1 (*strongly agree*) to 6 (*strongly disagree*). The PSOC has demonstrated good internal consistency (Cronbach's  $\alpha = 0.82$  for value/comfort, 0.70 for skills/knowledge) and strong test-retest reliability over 6 weeks ranging from 0.46 to 0.82. In the current study the PSOC demonstrated strong internal consistency (Cronbach's  $\alpha = 0.95$ ). The Hebrew version of the PSOC was implemented and has been validated in a previous study (Ophir et al. 2023).

#### 2.5.4 | Parental Depressive Symptoms

Parental depressive symptoms were measured using the Hebrew version of the Center for Epidemiologic Studies Depression Scale (CESD; Radloff 1977). This 20-item self-report questionnaire is designed to assess depressive symptomatology in the general population. The Hebrew version has demonstrated strong reliability and validity in Israeli samples (Bjorck and Lazar 2011). Participants rated the frequency of symptoms experienced over the past week on a 4-point Likert scale ranging from 0 (*rarely/none of the time*) to 3 (*most/all of the time*). Total scores result anywhere from 0 to 60 with higher scores reflecting greater levels of depressive symptoms. In the present study, the scale was used to examine changes in depressive symptoms over time and across treatment conditions while demonstrating a solid internal consistency (Cronbach's  $\alpha = 0.95$ ).

#### 2.5.5 | Adaptive Functioning

Children's adaptive functioning was assessed using the Adaptive Behavior Assessment System (ABAS; Harrison and Oakland 2015). Caregivers fill it out on behalf of children. ABAS measures age-normed adaptive skills across multiple arenas: a general adaptive composite (GAC) score as well as domain scores in conceptual, social, and practical functioning. The Hebrew version of the ABAS has been previously validated (Mecca et al. 2024). We focused on five indicators: the unstandardized composite score, the standard GAC score, the conceptual domain score, and two subdomains: communication and help. Parents completed the ABAS at two time points: baseline (T1) and 4 weeks later (T2). The second assessment followed completion of the four-session reflective functioning workshop for parents in the intervention group. For the waitlist control group, the second assessment occurred after an equivalent waiting period.

### 3 | Results

The primary aim of this study was to evaluate the efficacy of the intervention by comparing an intervention group with a waitlist control group pre- (T1) and immediately postintervention (T2). To examine potential long-term effects, additional follow-up assessments were conducted solely for the intervention group at 3 months (T3) and 6 months (T4) postintervention. The control group was not included as they received the intervention after the waitlist period. The intervention program was randomly divided into two groups equal in size ( $N1 \cong N2 = 125$ ). Background characteristics like age, parent's and child's gender, income, and education were compared across groups to validate the randomness of the group allocation. The trial ended once research goals were met; the trial yielded a natural stop.

Across the full program (34 workshops delivered over nearly 4 years from January 2021 to January 2025), recruitment and assessment procedures followed a consistent structure: approximately 2 weeks before each workshop, parents completed a semistructured interview and questionnaires followed by four weekly workshop sessions, a postintervention interview and questionnaires 2 weeks after the final session. Follow-up assessments occurred at 3 and 6 months after each workshop. For illustration, Workshop 10 began on February 6, 2023, and ended on February 27, 2023; follow-up assessments were subsequently conducted at 3 months (May 27, 2023) and 6 months (August 27, 2023). Workshop 34, the final workshop in the entire program, began on July 3, 2024, and ended on July 24, 2024, with its six-month follow-up completed in January 2025, marking the final follow-up assessment of the entire study. Due to incomplete records for the earliest workshops, the exact start date of Workshop 1 could not be verified (the consent form was signed on January 21, 2021).

#### 3.1 | Workshop Assessment Over Time

To evaluate the effectiveness of the intervention, three key indicators were assessed over time at four distinct time points—prior to the intervention (T1), immediately after (T2), 3 months postintervention (T3), and 6 months postintervention (T4). Two indicators, belief related to emotions (ITE) and PSE, were measured at all four time points. The third indicator, participants' perceived effectiveness of the workshop, was assessed at T3 and T4. A GEE analysis revealed a statistically significant effect of time on PSE (Wald  $\chi^2 = 17.81$ ,  $p < 0.001$ ) and for the perceived contribution of the workshop (Wald  $\chi^2 = 8.25$ ,  $p < 0.01$ ), with a marginally significant trend for beliefs related to emotions (Wald  $\chi^2 = 7.48$ ,  $p = 0.058$ ). Robust standard errors were estimated within the GEE procedure. Effect sizes were irrelevant as indicators (IVs) were binary (time and intervention) and were ranked based on post hoc pairwise comparisons for cases in which the main effect or the interaction were found to be significant.

Findings indicate a gradual improvement in malleability in beliefs related to emotions over time with mean scores decreasing from 2.24 (SE = 0.07) at baseline (T1) to 2.04 (SE = 0.10) at 6 months postintervention (T4). This suggests an improvement in participants' perceived emotional control and reflects improved ER. For PSE, mean scores increased from 5.07 (SE = 0.08) at T1 to 5.24 (SE = 0.07) at T2 and peaked at 5.33 (SE = 0.09) at T3. However, by T4, self-efficacy slightly declined to 5.22 (SE = 0.11). As this level was not significantly different from baseline, it indicates a temporary improvement that tapered over time. Regarding participants' perception of the workshop's effectiveness, scores increased over time from an average of 6.02 (SE = 0.29) at the 3-month follow-up (T3) to 6.57 (SE = 0.27) at the 6-month follow-up (T4). This suggests that participants increasingly recognized the workshop's benefits over time, possibly as they had more opportunities to apply its content in their everyday lives.

#### 3.2 | Parents' Emotions and Cognitive Indicators

Table 2 shows the time effects, group main effects, and their interaction effects on parental emotional and cognitive indicators. Effects are followed by the predicted marginal means. Overall

**TABLE 2** | GEE results for various types of parental feelings.

|                    | CESD                         | ITE                          | PSOC                            |                             |
|--------------------|------------------------------|------------------------------|---------------------------------|-----------------------------|
|                    | Last week negative emotions  | Emotion regulation beliefs   | Parental self-efficacy          | PDI                         |
| Main effects group |                              |                              |                                 |                             |
| Wald               | 0.81; 0.05                   | 0.53; 0.04                   | 1.75; 0.07                      | 0.001; 0.002                |
| Control            | 1.92<br>(0.05)               | 2.23<br>(0.06)               | 5.28<br>(0.07)                  | 4.62<br>(0.11)              |
| Intervention       | 1.98<br>(0.05)               | 2.17<br>(0.06)               | 5.14<br>(0.07)                  | 4.63<br>(0.13)              |
| Time               |                              |                              |                                 |                             |
| Wald               | 1.29; 0.07                   | 2.77 <sup>~</sup> ; 0.09     | 1.53; 0.07                      | 0.04; 0.01                  |
| Time 1             | 1.97<br>(0.03)               | 2.24<br>(0.05)               | 5.18<br>(0.06)                  | 4.63<br>(0.09)              |
| Time 2             | 1.94<br>(0.04)               | 2.16<br>(0.05)               | 5.23<br>(0.05)                  | 4.61<br>(0.10)              |
| 2-Way interaction  |                              |                              |                                 |                             |
| Group×time         | 3.26 <sup>~</sup> ; 0.11     | 2.76 <sup>~</sup> ; 0.10     | <b>10.47<sup>**</sup>; 0.19</b> | 9.75 <sup>**</sup> ; 0.18   |
| Control T1         | 1.96 <sup>b</sup><br>(0.05)  | 2.24 <sup>ab</sup><br>(0.06) | 5.30 <sup>b</sup><br>(0.08)     | 4.75 <sup>b</sup><br>(0.12) |
| Control T2         | 1.88 <sup>a</sup><br>(0.05)  | 2.23 <sup>ab</sup><br>(0.08) | 5.24 <sup>ab</sup><br>(0.07)    | 4.47 <sup>a</sup><br>(0.12) |
| Intervention T1    | 1.98 <sup>ab</sup><br>(0.05) | 2.25 <sup>b</sup><br>(0.07)  | 5.07 <sup>a</sup><br>(0.08)     | 4.51 <sup>a</sup><br>(0.14) |
| Intervention T2    | 2.00 <sup>ab</sup><br>(0.05) | 2.07 <sup>a</sup><br>(0.07)  | 5.24 <sup>b</sup><br>(0.07)     | 4.81 <sup>b</sup><br>(0.16) |

Note: Standard errors in parentheses. Latin letters for marginal mean ranking, “a” for the lowest and upward, based on multiple pairwise comparisons. Cramer’s  $V$  as effect size next to the Wald’s  $\chi^2(1)$ : 0.10 low; 0.30 medium; 0.50 large. Bold Wald’s  $\chi^2$  for post hoc significant marginal mean differences. <sup>a,b,c</sup>post-hoc ranking based on pairwise multiple comparisons. The Bonferroni-corrected  $p < 0.05$  was applied to indicate a paired difference.

<sup>\*\*</sup> $p < 0.01$ .

<sup>~</sup> $p < 0.10$ .

group differences beyond time, and time differences beyond group, i.e., main effects, were found to be insignificant ( $p > 0.05$ ). This means that changes over time could differ between groups and vice versa. The time by group interaction of the last week’s negative emotion measures (Wald’s  $\chi^2 = 3.26$ ,  $p < 0.10$ ) was decomposed and ranked.

The intervention group experienced increased negative feelings but it was not statistically significant; whereas the control group demonstrated a decrease in level (a vs. b). We found the time by group interaction exerted an effect on emotional regulation beliefs (ITE: Wald’s  $\chi^2 = 2.76$ ,  $p < 0.10$ ). Parents in the intervention group experienced, on average, a greater belief in their ability to change their feelings in comparison to prior to the intervention; the control group demonstrated similar trends.

A clear increase in self-efficacy was predicted for the intervention group while the control group showed no change, although the latter’s mean PSE was higher in the first place. The PDI analysis resulted in time\*group interaction as well (Wald’s  $\chi^2 = 9.75$ ,  $p < 0.01$ ). Among the control group, the PDI dropped from T1 ( $M = 4.75$ ,  $SE = 0.12$ ) to T2 ( $M = 4.47$ ,  $SE = 0.12$ ), while the PDI mean level among the intervention group increased (T1:  $M = 4.51$ ,  $SE = 0.14$ ; T2:  $M = 4.61$ ,  $SE = 0.16$ ). These interaction

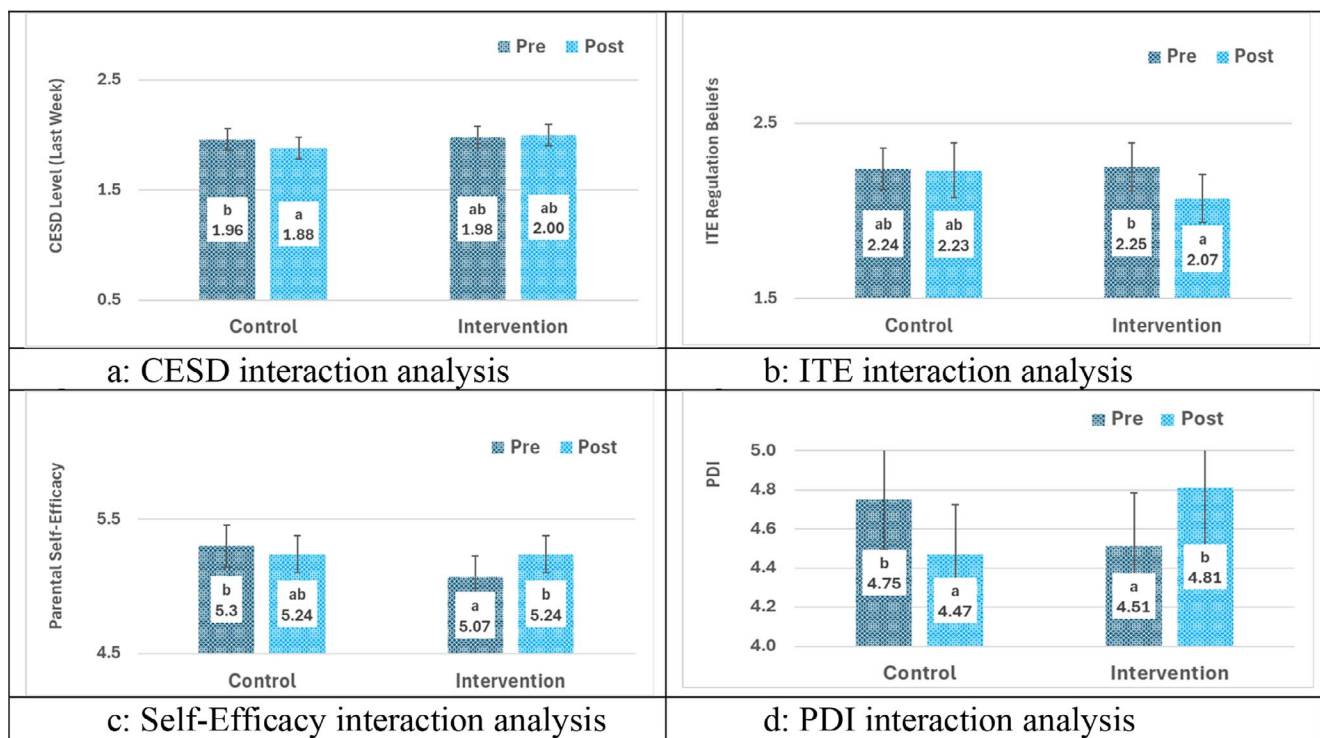
decompositions are shown in Figure 2a–d, where Latin letters represent the subgroup means from “a” upward as in Table 2.

### 3.3 | ABAS Indicator Analyses for Younger and Older Children

An additional data division was made for younger versus older children. Table 3 shows the analysis of four ABAS indicators: GAC; standard strategies (CON); social (SO); and practical (PR). The GEE analysis for the younger children resulted in time differences in the GAC and CON indicators where the levels at T2 were higher in comparison to T1 (not shown). The more interesting results among the younger children were found among the time × group interaction (GAC: Wald’s  $\chi^2 = 3.29$ ,  $p < 0.10$ ; PR: Wald’s  $\chi^2 = 3.29$ ,  $p < 0.01$ ). Among the intervention group, we found an increase in the mean GAC and PR, but these differences were insignificant among the control group. These differences are illustrated in Figure 3a,b. We could not find any differences between groups, times, or the interactions between the two effects among the older children (see Table 3).

Next, we looked at a broader and more specific set of ABAS indicators. Table 4 presents the results for younger (1–5 years, upper





**FIGURE 2** | Two-way interaction analyses by time and group. Figure (a) represents measures of parental depressive symptoms (CESD interaction analysis); (b) represents parents' implicit theories of emotion (ITE interaction analysis); (c) represents parental self-efficacy (self-efficacy interaction analysis); and (d) represents parental reflective functioning toward one's child (PDI interaction analysis).

part) and older (6–21 years, lower part) children. For the younger children, over time we found three ABAS indicators to vary on average regardless of the group effect (COM\_T: Wald's  $\chi^2=8.86$ ,  $p<0.01$ ; HS\_T: Wald's  $\chi^2=4.14$ ,  $p<0.05$ ; Wald's  $\chi^2=8.86$ ) where at T2 young children were found to have higher levels on average (not shown). Three interaction terms were found to be significant among the younger children (HS\_T: Wald's  $\chi^2=9.72$ ,  $p<0.01$ ; Sc\_T: Wald's  $\chi^2=7.53$ ,  $p<0.001$ ; SOC\_T: Wald's  $\chi^2=3.35$ ,  $p<0.10$ ). The change was observed among children from the intervention group for the first two indicators but no time differences were detected among the control group. The interaction decompositions for the Health and Help domains are presented in Figure 4a,b, respectively. The decomposition of the SOC\_T indicator did not result in a significantly different ranking. However, the predicted mean at T2 was lower for the control group and higher for the intervention group. The complete analyses of the ABAS indicators among the older children resulted in one interaction effect that was found to be significant (COM\_T: Wald's  $\chi^2=4.34$ ,  $p<0.05$ ). However, post hoc interaction decomposition did not determine a clear ranking although the control group mean was slightly lower at T2 whereas the intervention group mean was higher at T2. Figures 3 and 4 show these differences when they appear significant.

## 4 | Discussion

Our findings contribute to the growing literature on parental mental health and reflective capacity in the context of neurodiverse caregiving by identifying PRF as a central mechanism of change that supports parental well-being and child-related outcomes. PRF improved significantly in the intervention group, affirming the program's success. While PRF is a cornerstone of

sensitive and attuned parenting (Fonagy et al. 2002), parents of autistic children may struggle. Differences among autistic children in social communication, affective signaling, and behavioral rigidity can obscure cues typically used to infer internal states among nonautistic children, rendering PRF more effortful and emotionally taxing. As Slade (2009) emphasized, it is precisely autistic children who most depend on their caregivers' ability to mediate and scaffold their understanding of the social world.

Improvement in PRF was closely linked to increases in PSE, a robust predictor of parental adjustment and child functioning (Brennan et al. 2024; Zhou et al. 2018). By equipping parents with a framework to interpret their child's behaviors as meaningful, the intervention likely fostered a greater sense of competence, purpose, and agency. This finding is particularly important given the high rates of chronic stress, burnout, and helplessness reported by parents of autistic children (Hamama 2022). Our results suggest that changes in internal representations, rather than behavioral training alone, may play a powerful role in improving parental outcomes. Yet PSE tapered slightly by 6 months' time. Although still above baseline, this decline suggests that booster sessions or ongoing support may be needed to maintain long-term gains.

This study also assessed parents' beliefs about emotion malleability, a dimension rarely explored in autism intervention research, using the ITE scale. Results revealed that parents' ITE improved gradually over time, reflecting an increased confidence in their capacity for ER. This finding aligns with Tamir et al. (2007) and Kim et al. (2024), whose research supports the relationship between emotion malleability beliefs and better regulatory outcomes. Among participants, this shift may reflect a growing

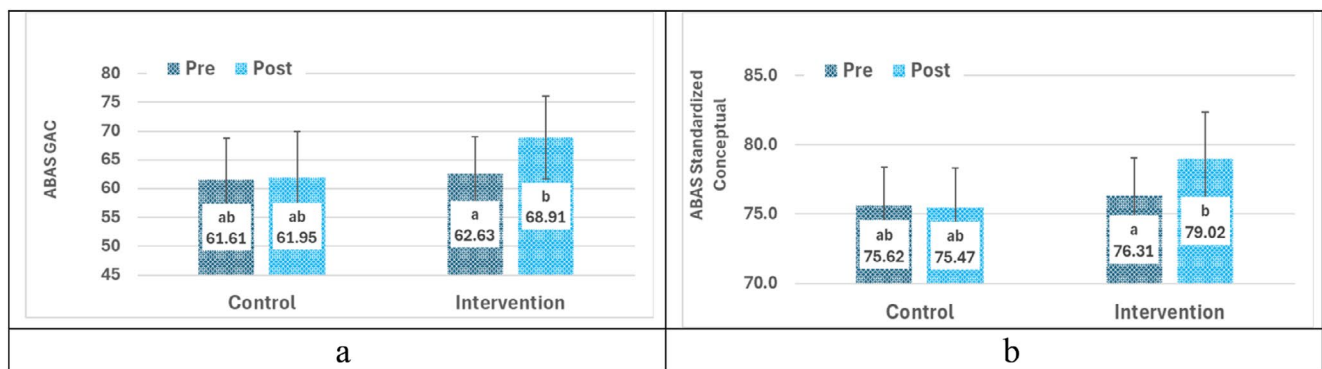
**TABLE 3** | GEE results for ABAS general indicators by time and intervention for children's age 1–5 years (upper) and 6–21 years (lower).

|                     | General adaptive composite (GAC) | Standard types of strategies (CON) | Social (SO)     | Practical (PR)                |
|---------------------|----------------------------------|------------------------------------|-----------------|-------------------------------|
| 2-Way interaction   |                                  |                                    |                 |                               |
| Children 1–5 years  |                                  |                                    |                 |                               |
| Group × time        | 3.29 <sup>~</sup> ; 0.11         | 0.85; 0.05                         | 2.07; 0.08      | 7.80 <sup>**</sup> ; 0.16     |
| Control T1          | 61.61 <sup>ab</sup><br>(3.66)    | 19.71<br>(1.37)                    | 11.77<br>(0.80) | 23.35 <sup>ab</sup><br>(1.43) |
| Control T2          | 61.95 <sup>ab</sup><br>(4.06)    | 20.56<br>(1.59)                    | 11.63<br>(0.91) | 22.75 <sup>ab</sup><br>(1.60) |
| Intervention T1     | 62.63 <sup>a</sup><br>(3.26)     | 20.42<br>(1.04)                    | 11.58<br>(0.79) | 22.81 <sup>a</sup><br>(1.36)  |
| Intervention T2     | 68.91 <sup>b</sup><br>(3.70)     | 22.51<br>(1.39)                    | 12.74<br>(0.99) | 25.61 <sup>b</sup><br>(1.40)  |
| Children 6–21 years |                                  |                                    |                 |                               |
| Group × time        | 0.95; 0.06                       | 2.27; 0.09                         | 0.02; 0.01      | 0.30; 0.03                    |
| Control T1          | 47.75<br>(2.50)                  | 16.48<br>(0.87)                    | 10.80<br>(0.64) | 20.30<br>(1.21)               |
| Control T2          | 46.44<br>(2.17)                  | 15.96<br>(0.83)                    | 10.85<br>(0.55) | 19.48<br>(1.04)               |
| Intervention T1     | 47.01<br>(2.67)                  | 16.44<br>(0.93)                    | 10.57<br>(0.67) | 20.01<br>(1.32)               |
| Intervention T2     | 48.68<br>(3.17)                  | 17.62<br>(1.13)                    | 10.71<br>(0.77) | 20.04<br>(1.54)               |

Note: Standard errors in parentheses. Cramer's  $V$  as effect size next to the Wald's  $\chi^2(1)$ : 0.10 low; 0.30 medium; 0.50 large. <sup>a,b,c</sup>post-hoc ranking based on pairwise multiple comparisons. The Bonferroni-corrected  $p < 0.05$  was applied to indicate a paired difference.

<sup>\*\*</sup> $p < 0.01$ .

<sup>~</sup> $p < 0.10$ .

**FIGURE 3** | Interaction decomposition of ABAS indicators. (a) ABAS GAC (general adaptive composite) for ages 1–5 and (b) ABAS PR (Practical) for ages 1–5.

confidence that emotional challenges can be addressed through reflective practice and relational support. It also suggests that change in beliefs related to emotions may serve as an additional mechanism of impact, reinforcing both PRF and emotional resilience. These findings are imperative given gains associated with parental variables (e.g., PRF, PSE, ITE) are closely linked to the child's gains in adaptive functioning.

Despite these gains, some parents in the intervention group reported a slight increase in negative emotional experiences following the program. This pattern may reflect the emotional

cost associated with greater awareness of and engagement with the complexities of caregiving. As parents deepen their reflective capacity, they may become more attuned not only to their child's inner world but also to past relational ruptures, present limitations, and the ongoing emotional demands of parenting an autistic child (Hestbaek et al. 2025). Increased reflective awareness may therefore initially heighten emotional sensitivity before contributing to longer-term adaptation and regulation.

As measured by parent-reported ABAS scores, child outcomes demonstrated an overall trend of improvement. Notably,

**TABLE 4** | GEE results for ABAS dimensions by intervention and time for age 1–5 years (upper) and age 6–21 years (lower).

|                     | ABAS<br>Com_T<br>Comm1      | ABAS<br>CU_T<br>Comm2 | ABAS<br>FA_T<br>Academic | ABAS<br>HL_T<br>Daily | ABAS<br>HS_T<br>Health      | ABAS<br>LS+T<br>Leisure | ABAS<br>Sc_T<br>Help          | ABAS<br>SD_T<br>Directive | ABAS<br>SOC_T<br>Social |
|---------------------|-----------------------------|-----------------------|--------------------------|-----------------------|-----------------------------|-------------------------|-------------------------------|---------------------------|-------------------------|
| Children 1–5 years  |                             |                       |                          |                       |                             |                         |                               |                           |                         |
| Group × time        | 0.55;<br>0.04               | 0.01;<br>0.01         | 0.72;<br>0.05            | 2.55;<br>0.09         | 9.72**;<br>0.18             | 0.55;<br>0.04           | 7.53** <sup>a</sup> ;<br>0.16 | 0.23;<br>0.03             | 3.35;<br>0.11           |
| Control T1          | 5.97<br>(0.50)              | 6.35<br>(0.48)        | 7.87<br>(0.65)           | 6.39<br>(0.35)        | 5.26 <sup>a</sup><br>(0.40) | 6.52<br>(0.53)          | 5.35 <sup>b</sup><br>(0.51)   | 5.87<br>(0.47)            | 5.26<br>(0.38)          |
| Control T2          | 6.41<br>(0.62)              | 6.71<br>(0.54)        | 7.76<br>(0.79)           | 6.24<br>(0.41)        | 5.10 <sup>a</sup><br>(0.43) | 6.72<br>(0.59)          | 4.73 <sup>a</sup><br>(0.47)   | 6.40<br>(0.51)            | 4.91<br>(0.39)          |
| Intervention T1     | 6.59<br>(0.43)              | 6.49<br>(0.48)        | 8.56<br>(0.56)           | 6.16<br>(0.41)        | 5.71 <sup>a</sup><br>(0.44) | 6.57<br>(0.46)          | 4.43 <sup>a</sup><br>(0.33)   | 5.30<br>(0.47)            | 5.01<br>(0.45)          |
| Intervention T2     | 7.43<br>(0.44)              | 6.89<br>(0.46)        | 9.00<br>(0.61)           | 6.81<br>(0.44)        | 6.72 <sup>b</sup><br>(0.49) | 7.21<br>(0.52)          | 5.24 <sup>b</sup><br>(0.45)   | 6.11<br>(0.64)            | 5.54<br>(0.57)          |
| Children 6–21 years |                             |                       |                          |                       |                             |                         |                               |                           |                         |
| Group X Time        | 4.34*;<br>0.12              | 0.63;<br>0.05         | 1.72;<br>0.08            | 0.03;<br>0.01         | 0.07;<br>0.02               | 0.37;<br>0.04           | 0.13;<br>0.02                 | 0.30;<br>0.03             | 0.74;<br>0.05           |
| Control T1          | 6.09 <sup>a</sup><br>(0.34) | 4.85<br>(0.38)        | 5.91<br>(0.33)           | 4.37<br>(0.38)        | 6.55<br>(0.40)              | 6.31<br>(0.33)          | 4.54<br>(0.32)                | 4.48<br>(0.34)            | 4.49<br>(0.37)          |
| Control T2          | 5.78 <sup>a</sup><br>(0.31) | 4.46<br>(0.37)        | 5.71<br>(0.34)           | 4.34<br>(0.36)        | 6.21<br>(0.36)              | 6.52<br>(0.30)          | 4.47<br>(0.29)                | 4.48<br>(0.34)            | 4.33<br>(0.31)          |
| Intervention T1     | 5.92 <sup>a</sup><br>(0.34) | 5.31<br>(0.45)        | 5.98<br>(0.37)           | 3.96<br>(0.37)        | 6.01<br>(0.45)              | 6.33<br>(0.34)          | 4.64<br>(0.36)                | 4.47<br>(0.39)            | 4.23<br>(0.38)          |
| Intervention T2     | 6.36 <sup>a</sup><br>(0.38) | 5.33<br>(0.55)        | 6.40<br>(0.46)           | 4.02<br>(0.46)        | 5.82<br>(0.48)              | 6.30<br>(0.38)          | 4.70<br>(0.40)                | 4.75<br>(0.50)            | 4.42<br>(0.43)          |

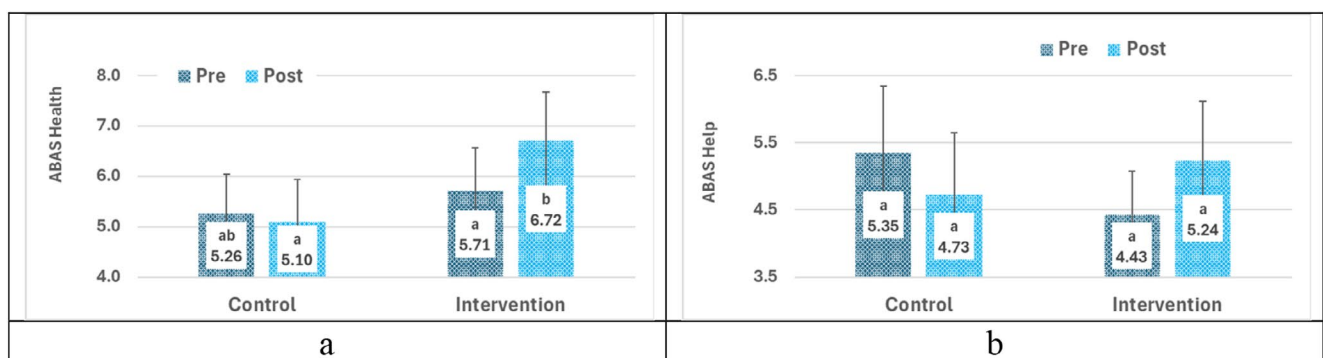
Note: Standard errors in parentheses. Cramer's  $V$  as effect size next to the Wald's  $\chi^2(1)$ : 0.10 low; 0.30 medium; 0.50 large. <sup>a,b,c</sup>post-hoc ranking based on pairwise multiple comparisons. The Bonferroni-corrected  $p < 0.05$  was applied to indicate a paired difference.

Abbreviations: Comm1, communication; Comm2, community.

\* $p < 0.06$ .

\*\* $p < 0.01$ .

<sup>†</sup> $p < 0.10$ .



**FIGURE 4** | Interaction decomposition of ABAS dimensions. (a) ABAS health for age 1–5 and (b) ABAS help for age 1–5.

significant gains were observed primarily among younger children, particularly in the communication and conceptual domains, whereas comparable changes were not detected among older children. These findings may align with developmental models emphasizing heightened developmental plasticity in early childhood (Fandakova and Hartley 2020), suggesting that relational interventions delivered during earlier developmental

periods may exert stronger effects on parent–child interaction patterns and adaptive functioning. Early childhood may represent a particularly sensitive developmental stage in which changes in caregiving processes, emotional attunement, and co-regulation more readily shape children's emerging adaptive and communicative capacities. At the same time, autistic neurodevelopment may follow distinct trajectories of plasticity and

adaptation (Chen et al. 2024), underscoring the need for further research examining developmental timing and intervention responsiveness across age groups.

Several additional explanations may account for the differential findings between younger and older children. First, developmental differences may influence how parents perceive and report behavioral change (Ehrlich et al. 2011). Parents are often more directly involved in the daily emotional and adaptive functioning of younger children and may therefore be more likely to observe subtle shifts in communication, regulation, and interaction patterns. In contrast, behavioral and emotional changes in older children may be less visible within everyday parent-child interactions. In addition, the ABAS may be more sensitive to developmental and relational changes during early childhood, a period characterized by rapid developmental progression and greater reliance on parental scaffolding (Grzadzinski et al. 2020). The version designed for older children may be less sensitive to subtle emotional or interactional changes associated with increases in PRF. Together, these findings underscore the importance of considering developmental stage and measurement sensitivity when evaluating intervention effects across age groups (Tassé et al. 2012).

The observed gains in the younger cohort highlight the potential impact of parental insight and relational change alone. Improvements in PRF may alter the emotional tone and quality of parent-child interactions by creating greater opportunities for co-regulation, emotional security, and learning during a formative stage of development. Interestingly, participants' perceived value of the workshop continued to increase over time, possibly reflecting the gradual integration of workshop principles into daily parenting routines and relationships. These findings provide encouraging preliminary evidence that even brief, short-term interventions may produce enduring relational and emotional effects, although future research is needed to determine how such gains can best be maintained over time. Future studies should also consider developmentally sensitive and age-appropriate outcome measures to better capture the full range of child-level impacts across developmental periods.

Importantly, the observed effects may extend beyond statistical significance alone and reflect functionally meaningful changes within emotionally demanding caregiving contexts. Improvements in PRF and PSE may indicate enhanced parental capacity to regulate emotional responses, sustain reflective engagement during stressful parent-child interactions, and respond to children in a more attuned, flexible, and emotionally responsive manner. Given the elevated emotional, interpersonal, and regulatory demands often experienced by parents of autistic children, even relatively modest improvements in these domains may meaningfully influence daily family functioning and the quality of parent-child relationships.

Moreover, while several observed effect sizes were small, prior work suggests that statistically modest effects may nevertheless carry meaningful practical implications when applied across broader populations and real-world caregiving settings (Carey et al. 2023). More broadly, the functional significance of changes in reflective caregiving processes may not be fully captured by statistical magnitude alone, particularly in chronically stressful

parenting contexts (Cuijpers 2021). At the same time, the present study was not designed to establish clinically significant change using diagnostic or cutoff-based criteria. Accordingly, interpretations regarding clinical significance should remain cautious; future research should further examine the long-term functional and clinical implications of these improvements using observational and multi-informant approaches across diverse family populations.

Further, the online format of the intervention presents an efficient model that can easily be employed to parents everywhere. The format boosts connectivity, creates greater accessibility, and a sense of community (Hermaszewska and Sin 2021). As the current intervention is based on the in-person pilot by Enav et al. (2019) and the former demonstrates the efficacy of an online format, the current intervention also presents an opportunity for further customization to fit other populations as needed. Participant pools from other geographic locations can benefit from understanding the mechanisms underlying these interventions, and the use of other measures less reliant on self-report can provide more objective data.

#### 4.1 | Study Limitations

This study has several limitations. First, aside from the PDI, outcomes relied on self-report tools, introducing potential response biases. For instance, as the study relied in part on parental reports of their children during interviews with RAs, an observer-expectancy effect may have emerged. Further bias may have been introduced into the feedback when parents reported on their children in relation to the ABAS due to biases in symptom perception (Smith et al. 2020). Additionally, the differing monetary incentives supplied to each group may have introduced potential bias as those who received more compensation may feel more inclined to follow through with the study. Future work should include multi-method, multi-informant assessments and keep any incentives, monetary or otherwise, balanced across groups.

Second, recruitment was voluntary so parents who chose to participate may have been more self-aware, motivated, or possessed greater resources (e.g., higher digital literacy) which may limit external validity. Third, the trial was conducted in Israel and delivered entirely online, constraining generalizability to other cultural contexts and to families without reliable internet access. Fourth, the intervention and follow-up were brief, and T3/T4 assessments were obtained only for the intervention arm as the control functioned as a waitlist due to ethical considerations.

Fifth, in addition, although the overall sample size was adequately powered for the primary analyses of the full randomized controlled trial sample, the study was not specifically powered for age-stratified subgroup comparisons. Accordingly, findings related to developmental differences between younger and older children should be interpreted cautiously and considered exploratory pending replication in larger samples. In addition, several marginal interaction effects did not remain significant following more conservative corrections for multiple comparisons and should therefore be interpreted as exploratory.



Further, the current study's design may be strengthened through the inclusion of an active control group. This group may be provided psychoeducational materials or an alternative intervention to better compare the statistical significance of the intervention's results. We encourage future research to implement this strategy to build upon the presented findings. Future studies should also incorporate longer follow-up, equivalent time points for both arms, and where feasible, an active or classic control condition to better isolate treatment effects.

## 4.2 | Clinical Implications

This RCT delivered a complete, brief, fully online program to a vulnerable population, demonstrating feasibility at scale. The virtual, group-based format expands access for families living far from services, during periods of restricted mobility (e.g., pandemics), and for caregivers who find leaving home difficult due to childcare, transportation, or health constraints. Scalability of the intervention is entirely viable due to its online nature. However, the challenge with expanding on its scope is the cost associated with human resources—the current intervention requires trained mental health professionals who attend supervision sessions and discuss findings as a team. Further standardization of the intervention such as structured training sessions hosted on online platforms and group supervision may lead to easier scaling of the intervention. Financial challenges associated with facilitation and supervision may be mitigated through larger training and supervision groups. Further, even though the online format allowed for participants to attend regardless of geographic location, participants still dropped out primarily due to time constraints. It may be necessary to offer several intervention workshops across different dates and times to better serve the needs of parents of autistic children.

The trial underscores the central role of PRF in support for parents of autistic children. Embedding PRF elements like curiosity about the child's inner world, emotional flexibility, and perspective taking within parenting programs (see Huynh et al. 2024) may help parents regulate their own emotions, build resilience, and foster more adaptive parent-child interactions. Findings also suggest that strengthening PRF can indirectly enhance parenting self-efficacy and perceived child behavior, particularly for younger children. Incorporating brief PRF screening could enable more tailored, precision-focused interventions, optimizing engagement and outcomes in high-stress caregiving contexts.

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## Ethics Statement

This study obtained approval from the IRB of University of Haifa protocol number 60124.

## Consent

All participants provided their informed consent prior to participation.

## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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